

Two new exact large- k values for sets of unit vectors with small subset sums

Candidate partial solution of Conjecture 28 in arXiv:1210.0366

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Status. Candidate proof of two previously omitted cases of Conjecture 28; substantial partial result. Likely valid, novelty uncertain; send to human review.

1 Source problem and result

For $k, d \geq 2$, let $\bar{\mathcal{C}}(k, d)$ be the largest size, over all real d -dimensional Banach spaces X , of a family of unit vectors $x_1, \dots, x_m \in X$ such that

$$\left\| \sum_{i \in I} x_i \right\| \leq 1 \quad \text{for every } I \subseteq \{1, \dots, m\}, \quad |I| = k.$$

Swanepoel's Theorem 26 gives $\bar{\mathcal{C}}(k, 6) = k + 1$ for $k \geq 17$ and $\bar{\mathcal{C}}(k, 7) = k + 1$ for $k \geq 41$. Conjecture 28 predicts the same conclusion whenever $k \geq 2d - 1$.

(2) If $-2d + \sqrt{6d^2 + 3d + 1} \leq k \leq 2d - \sqrt{d/2}$ then $\mathcal{C}(k, d) = 2d$.

(3) If $d \geq 3$ and $k > 2d - \sqrt{d/2}$ then $\bar{\mathcal{C}}(k, d) \leq k + \frac{1 + \sqrt{2d-3}}{2}$.

For values of d up to 7 as $k \rightarrow \infty$ the same methods as used in proving Theorems 20, 23, 24 and 25 give the following exact values.

Theorem 26. $\bar{\mathcal{C}}(k, d) = \max \{k + 1, 2d\}$ in the following cases:

- (1) $d = 2$ and $k \geq 2$,
- (2) $d \in \{3, 4, 5\}$ and $k \geq 3$,
- (3) $d = 6$ and $k \in \{3, 4, 5, \dots, 10\} \cup \{17, 18, 19, \dots\}$,
- (4) $d = 7$ and $k \in \{3, 4, 5, \dots, 12\} \cup \{41, 42, 43, \dots\}$.

The proof method gives no information for $d \geq 8$ and k large. (The estimate $\bar{\mathcal{C}}(2, 3) \leq 9$ is also obtained in the proof.) For arbitrary d , as long as k is large, we obtain the following using a completely different technique.

Theorem 27. If $k \gg d^{d+2}$ then $\bar{\mathcal{C}}(k, d) = k + 1$.

The proof uses geometric tools from convexity, in particular the Brunn-Minkowski inequality and the theorem of Carathéodory. The hypothesis $k \gg d^{d+2}$ is most likely not best possible, but we need at least $k \geq 2d - 1$ for the conclusion of this theorem to hold, as shown by the example of $k \leq 2d - 2$ and the family $\{\pm \mathbf{e}_i \mid i \in [d]\}$ in ℓ_∞^d .

Conjecture 28. $\bar{\mathcal{C}}(k, d) = k + 1$ whenever $k \geq 2d - 1$.

Theorem 1. The two endpoint values immediately below the source's displayed ranges are

$$\boxed{\bar{\mathcal{C}}(16, 6) = 17}, \quad \boxed{\bar{\mathcal{C}}(40, 7) = 41}.$$

The proof keeps the integer jump position in the source's scalar optimization. At precisely these parameters the resulting Frobenius bound forces equality in the rank inequality, and its equality structure is impossible.

2 A sharp scalar lemma at the two endpoints

Call $\alpha = (\alpha_1, \dots, \alpha_m)$ *k-collapsing* if $|\sum_{i \in I} \alpha_i| \leq 1$ for every k -subset I .

Lemma 1. *Suppose $k = m - 2$, $\alpha_m = 1$, and α is k -collapsing.*

1. *If $m = 18$, then $\sum_{j=1}^{18} \alpha_j^2 \leq 3$. Equality holds, up to permuting the first seventeen coordinates, only for*

$$(\alpha_1, \dots, \alpha_{17}) = (\underbrace{1/2, \dots, 1/2}_5, \underbrace{-1/4, \dots, -1/4}_{12})$$

or

$$(\alpha_1, \dots, \alpha_{17}) = (\underbrace{3/7, \dots, 3/7}_6, \underbrace{-2/7, \dots, -2/7}_{11}).$$

The corresponding complete row sums are $1/2$ and $3/7$.

2. *If $m = 42$, then $\sum_{j=1}^{42} \alpha_j^2 \leq 6$. Equality holds, up to permutation, only when the first forty-one coordinates consist of thirteen copies of $1/2$ and twenty-eight copies of $-1/4$. The complete row sum is $1/2$.*

Proof. First note the elementary normalization $|\alpha_j| \leq 1$ for $j \neq m$. Indeed, comparing two k -subsets differing only by m and j gives $\alpha_j \geq -1$. There are at most $k - 1$ positive coordinates, since a k -subset of positive coordinates containing $\alpha_m = 1$ would have sum greater than one. As $m - k = 2$, choose two nonpositive coordinates other than j, m and compare a k -subset containing j, m with one in which those two coordinates replace j, m . The upper and lower subset bounds give $\alpha_j \leq 1$.

Write the remaining coordinates in decreasing order $x_1 \geq \dots \geq x_{m-1}$. A k -subset either includes the distinguished entry 1 and omits two of the x_j , or excludes 1 and omits one x_j . Using $|x_j| \leq 1$ and the ordering, all the subset inequalities are equivalent to

$$x_1 + \dots + x_{m-3} \leq 0, \quad x_2 + \dots + x_{m-1} \geq -1. \quad (1)$$

For example, the two displayed inequalities are the upper bound for the largest sum including 1 and the lower bound for the smallest sum excluding 1. They imply the other two extreme bounds because $x_{m-2} \leq 0$ and $x_2 \leq 1$.

We maximize the strictly convex function $Q(x) = \sum_j x_j^2$ over the polytope given by the ordering and (1). An elementary active-constraint calculation lists its vertices. If all coordinates agree, they are either all zero or all $-1/(m-2)$. Otherwise there is one jump: for some $1 \leq t \leq m-3$, the first t coordinates equal

$$a_t = \frac{m-3-t}{t+m-3}, \quad b_t = \frac{-t}{t+m-3}$$

and the remaining $m-1-t$ coordinates equal b_t . The last vertex is $(0, \dots, 0, -1)$. This list follows because at a nonconstant vertex both inequalities in (1) are active and all but one of the ordering inequalities are active. Conversely, the displayed points meet the constraints.

At a one-jump vertex the off-diagonal square sum is

$$q_m(t) = ta_t^2 + (m-1-t)b_t^2 = \frac{t((m-3)^2 - (m-5)t)}{(t+m-3)^2}. \quad (2)$$

As a function of real t , its derivative has the sign of

$$(m-3)^2 - (3m-13)t.$$

Thus it is unimodal, with real maximizer $(m-3)^2/(3m-13)$.

For $m = 18$ this maximizer is $225/41 \in (5, 6)$. Direct substitution gives $q_{18}(5) = q_{18}(6) = 2$, while unimodality and the remaining vertices show that these are the only maxima. The formulas for a_t, b_t give the two stated rows and their sums. For $m = 42$ the real maximizer is $1521/113 \in (13, 14)$; here

$$q_{42}(13) = 5, \quad q_{42}(14) = \frac{14042}{2809} < 5.$$

Hence $t = 13$ is the unique integer maximizer, giving the stated row. Adding the distinguished square $\alpha_m^2 = 1$ proves both bounds and equality statements. $\square$

3 Rank equality rules out both families

We use the sharp trace–Frobenius inequality

$$|\operatorname{tr} A|^2 \leq \operatorname{rank}(A) \sum_{i,j} |a_{ij}|^2. \quad (3)$$

For a real matrix, equality holds exactly when A is symmetric and all its nonzero eigenvalues are equal. For completeness, this follows by applying Schur triangularization, Cauchy–Schwarz to the nonzero eigenvalues, and unitary invariance of the Frobenius norm. Equality makes the Schur form diagonal with equal real nonzero diagonal entries.

Proof of Theorem 1. Suppose first that eighteen unit vectors in a six-dimensional Banach space are 16-collapsing. Choose a norming functional f_i for each x_i and form

$$A = (a_{ij})_{i,j=1}^{18}, \quad a_{ij} = f_i(x_j).$$

Then $\operatorname{rank} A \leq 6$, $a_{ii} = 1$, and every row is a scalar 16-collapsing family. Lemma 1 and (3) give

$$6 \geq \operatorname{rank} A \geq \frac{18^2}{18 \cdot 3} = 6.$$

Every inequality is therefore an equality. In particular, A is symmetric, has rank six, and all its nonzero eigenvalues equal $\operatorname{tr} A/6 = 3$. Every row is one of the two equality types in Lemma 1.

The types cannot mix. Indeed, an off-diagonal entry shared by rows of different types would have to belong simultaneously to $\{1/2, -1/4\}$ and $\{3/7, -2/7\}$, which are disjoint. Hence all rows have the same type, so the all-ones vector is an eigenvector of A with eigenvalue equal to the common row sum, either $1/2$ or $3/7$. This contradicts the fact that every nonzero eigenvalue is 3. Thus no such eighteen-vector family exists.

For forty-two unit vectors in dimension seven that are 40-collapsing, the same construction and the second part of Lemma 1 yield

$$7 \geq \operatorname{rank} A \geq \frac{42^2}{42 \cdot 6} = 7.$$

Equality makes A symmetric of rank seven with every nonzero eigenvalue equal to $42/7 = 6$. Every row has the unique equality type and row sum $1/2$, so the all-ones vector is instead an eigenvector with eigenvalue $1/2$, again a contradiction.

Any larger family contains a subfamily of the forbidden size, while preserving the fixed k -subset condition. Therefore $\bar{\mathcal{C}}(16, 6) \leq 17$ and $\bar{\mathcal{C}}(40, 7) \leq 41$. Conversely, in a two-dimensional subspace take $k+1$ unit vectors summing to zero (for example, a Euclidean regular $(k+1)$ -gon). Every k -sum is the negative of the omitted unit vector. Hence $\bar{\mathcal{C}}(k, d) \geq k+1$, proving both equalities. $\square$

4 Verification, scope, and novelty

Exact-rational enumeration independently checks the two maxima, equality positions, row sums, and rank thresholds. It is included only as a sanity check. The neighboring cases $(d, k) = (6, 15)$ and $(7, 39)$ merely force the rank to be at least the ambient dimension, not equality, so the spectral contradiction does not extend. A scan of the same endpoint formula through $m < 2000$ found no other exact rank-equality hits. The full Conjecture 28 remains open.

A bounded search through 11 August 2026 used the exact title and terminology and the citation graph of DOI 10.1090/tran/6601. It found a survey chapter and an adjacent signed-subset-sum paper, but no source stating either endpoint value. Priority is not asserted.

Reference. K. J. Swanepoel, *Sets of unit vectors with small subset sums*, arXiv:1210.0366v2; Trans. Amer. Math. Soc. 368 (2016), 7153–7188, especially Theorem 26, Conjecture 28, and the matrix/rank method in Sections 5 and 7.